A Predictive Analysis of lap time in Formula 1 using Machine Learning

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2025 IEEE 6th India Council International Subsections Conference (INDISCON) | 979-8-3315-1504-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/INDISCON66021.2025.11252332

ABSTRACT

Forecasting lap time in Formula 1 is challenging because of the nature of the data to have complex relationships between the features. This research paper discusses various machine learning models like Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression(GBR), K-Nearest Neighbor (KNN), and XGBoosting(Extreme Gradient Boosting) to forecast the lap times of races. The data were curated by removing the outliers in the data set and normalizing the data, improving the models' accuracy. Each model uniquely deals with the dataset, with Random Forest Regression giving the best results, having an R^2 score for the training data equal to 0.9969 and the R^2 score of the test data equal to 0.9927 while having low MSE and MAE. This shows the predictive potential of ensemble-based machine learning models, especially the Random Forest Regression, when making predictions on complex data sets such as that of Formula 1. This can have several applications and can benefit teams in building better strategies and performing performance analyses.

***Index Terms*—Formula 1, Lap Time, Machine Learning, Supervised learning, Algorithms, Regression Algorithms.**

I. INTRODUCTION

Formula 1 represents the pinnacle of motorsport, followed by millions of people worldwide. State-of-the-art machines go through high-speed tracks at record-breaking speeds. This sport depends not only on the driver's ability to drive but also on carefully constructed strategies to scrape every fraction of a second possible in order to be ahead of their competitors. Data plays a vital role in this highly competitive sport. Teams collect telemetry data from live runs and other data, including driver behavior patterns and track specifications, to build race strategies and try to fine-tune the car to the driver's behaviors and habits to scrape out more performance. One of the most important optimization strategies for this sport is to find the right interval for a pit stop, which requires highly accurate predictions of the lap times. In a race, three types of tire compounds are available: soft, medium, and hard, given by the official tier provider Pirelli. Soft tires have the highest amount of wear and grip, whereas hard tires compensate for speed but are highly durable. Depending on the data collected by the engineering team and the proper analysis of various characteristics, teams develop highly optimized pit stop strategies with precise execution to help the driver reduce lap time. Emerging improvements in the field of Machine learning models have made the data acquisition process and analysis of data for each team far more manageable. Teams have developed specialized algorithms to collect and analyze telemetry data during practice runs and also live data that comes in during the final race and make real-time changes to strategies for increasing optimizations such as lower pit stop times and better pit stop intervals, which could help the team to have a competitive edge over others. This increasing shift in Formula 1 to being highly data driven shows how boundaries in motorsport can be pushed further and how technology can increase the competitiveness of the sport. Advancements in technology with increasing applications of Machine Learning and Deep Learning in highly data-dependent sports are making new highs and paving new paths in F1. The primary contribution of this paper is the construction of a feature called 'LapTimeDelta'.This feature calculates the variance in a driver's lap time through a race through multiple laps. It is helpful in analyzing the consistency of a driver throughout a race, which is crucial data for creating personalized dynamic strategies for each driver and creating relations with other dynamic features like track temperature and corner entry speed.

Subsequent sections include a literature survey covering relevant topics. In the methodology section, the paper discusses the process of acquiring and handling the dataset used for the study.It outlines the sources of data, the methods employed for data collection, and the techniques applied for preprocessing , cleaning, and analysis which then moves on to talk about various models like Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression (GBR), K-Nearnest Neighbor(KNN) and XGBoost (Extreme Gradient Boosting) that are used for making lap-time predictions. Moving on to results and discussion, The results and discussion section presents a comparison of R², MAE, and MSE for various models and discuss why the Random Forest Regressor is the best model for this dataset, along with a mathematical analysis of it. The results and discussion section presents a comparison of R², MAE, and MSE for various models the future applications of such prediction models in Formula 1.

II. LITERATURE AND SURVEY

The author in [1] emphasizes the proper determination of the number of laps that best represents a tyer to develop dynamic base pit stop strategies built on practice data, which change based on various track factors and driving styles. Regression models are used to compare the pace of multiple tires and find the best one, followed by linear programming to develop multiple base cases for various strategies. Then, a neural network was used to predict pit stops in the race.

The author in [2] discusses the use and evaluation of supervised learning algorithms to predict the race winner and driver standings. The best qualifying time is the most critical factor in predicting the race winner which directly relates to the driver's position on the grid.

The author in [3] talks about timing and evaluates the correctness of pit stops using machine learning algorithms like SVM, RandomForest, and Neural Networks. The raw data is feature engineered, and two variables, 'has pit stop' and 'good pit stop' are made before training the models. SVM shows the best accuracy while predicting the variables 'has pit stop' and 'good pit stop' among the three models. The author also highlights the presence of overfitting and underfitting in the models.

In [4], the author highlights the use of deep learning and Neural Network algorithms to predict the tire strategy of a race. LSTM, GRU, and MLP are used to indicate the strategy here, and GRU performed the best among all of them, signifying the importance of continuity and long-term dependencies in a dataset to make valuable predictions.

The author of [5] tries to explain how the most important features to predict race outcomes are extracted from publicly available datasets. Using the weather data frame to make relations with the outcome was difficult because there were no standard features. Still, the most relevant target features were humidity, pressure, and air temperature. In the grid data frame, the average position in the grid, the breaking point, and the variation in the breaking point were essential to finding the final standings of a race.

The author in [6] shows the importance of generating accurate predictions through machine learning to enhance decision making in NASCAR cup races. Here, there is a use of software developed by MIT, which is currently running several machine learning models. The parameters that significantly impact the drivers' speed are the car's relative position on the grid car setup and the track characteristics of tire flags. Another external factor that had a significant impact was the driver's skills. In its application to find the momentum of track position gain or loss, it was concluded that a driver's momentum can be predicted early in a race; if a driver gains positions early in a race, then the driver continues to do so through the race, and it holds true for the opposite. This is an essential feature as it can be used to get better accuracy in decision-making.

In [7], the author's objective was to predict the 2021 Grand Prix Championship results using both Artificial Neural Networks and Linear Regression models on publicly available data to see which is more suitable for the Formula 1 data. The top three positions were predicted accurately by the Artificial Neural Network, whereas in Machine Learning Regression, the First and Second places were in-verted. However, The Machine Learning Regression is better at predicting the positions of the first 10 with an error of just one position, while Artificial Neural Networks showed relatively more significant deviations. After comparing the predictions of all the positions on the grid, it was concluded that Artificial Neural Networks are better at analyzing Formula 1 data.

In [8], the objective was to find if the track characteristics, the motorcycle specifications, and the driver details were sufficient to find the Championship results. Here, no particular model used is singly prevalent or considered the best performer, but it can be seen that Gradient Boosting performed the best at predicting podium places in MotoGP. When predicting the winners, Random Forest was the most well-performing model.

The author of [9] explores the integration of Artificial Neural Network models into Virtual Strategy engineering to make accurate pit-stop predictions. The VSE is trained specially for each race and particular drivers, and to make it applicable universally, it should be trained with random drivers before every training.

The author in [10] is concerned with improving drivers' performance with the help of linear regression models. The data of the driver's previous races are recorded and stored in a database. The data is then processed with basic statistics and linear regression. The data that is stored fits the Machine Learning model to find patterns. The found patterns are then used to give recommendations to the driver to improve their performance.

In the paper [11], the author focuses on how various machine learning models forecast lap times, sector performance, and race strategy. The dataset consisted of race data between 2018 and 2023 taken from the FastF1 library. Principal Component Analysis (PCA) and Recursive Feature Elimination (RFE) were implemented to find the most important features for making predictions. Models like Random Forest, Support Vector Machines (SVM), Gradient Boosting, and K-Means Clustering were used. After constructing models, it was found that the Gradient Boosting had the lowest Mean Squared Error (MSE) for lap time prediction and that it outperformed all the other models, with Tire compound and track temperature being the most important feature to predicting the lap times.

In [12] the author explores how deep learning model enhances perception, localization, decision making, and sensor fusion in autonomous vehicles. It highlights the placement of CNNs in object detection, LSTMs in motion prediction, and reinforcement, gaining knowledge of (RL) in driving approach optimization. The test discusses SLAM-based absolute localization integrating DeepVO and PoseNet for advanced positioning, while sensor fusion techniques combine LiDAR, radar, and cameras for advanced environmental awareness. Key annoying conditions embody immoderate computational demands, facts, annotation issues, model reliability, and real-worldwide deployment constraints. The paper emphasizes future research on lightweight AI models, self-supervised gaining knowledge of, adversarial training for safety validation, and neuromorphic computing for inexperienced processing, aiming to make deep gaining knowledge of models greater robust and deployable in real-time self enough driving systems.

The author of [13] explores the integration of Industry 4.0 technologies into Formula 1 and other motorsports. In the case study, the Mercedes-AMG Petronas F1 Team was used to create an optimized fuel and tire change strategy. Red Bull Racing was used for aerodynamic modeling, and Ferrari and McLaren were used as digital twins to compare simulation data. It was concluded that AI analytics improved race strategy efficiency by 12 percent.

The author in [14] focuses on design and implementation of an autonomous racing system. It describes AI-based perception, planning, and control systems, real-time decision-making, trajectory optimization, and competitor interaction. It addresses challenges such as high-speed perception, racing line optimization, vehicle stability, and multi-agent decision-making. The racing system consists of LiDAR, cameras, GPS, and radar for real time mapping, CNN-based vision pro-cessing for track detection and RNNs with LSTMs for competitor trajectory pre-diction. Model Predictive Control (MPC) optimizes motion planning, while Deep Reinforcement Learning (DRL) enables adaptive overtaking. Testing environments include simulations in CARLA and Gazebo and track tests at Indianapolis Motor Speedway. The research found that Lap times improved by 9.4 percent with AI-optimized racing lines, AI successfully completed overtakes in 93.2 percent of test cases without collisions, and Multi-agent AI predicted competitor actions up to 2.3 seconds in advance. These developments enhanced AI-driven race strategy and are applied in real- time autonomous driving, which includes merging and evasive maneuvering.

The author of [15] explores how the changes in the thermal and frictional properties of the tires cause changes in the performance of the vehicles. The test uses the TRT (Tire/Road Thermal) Model and GrETA (Grip Estimation for Tire Analysis) Model to analyze how reduced tread thickness affects various varia-bles like temperature distribution, friction components, and preferred grip through the tires. Findings propose that tire placement modifies the steadiness amongst adhesive and hysteretic friction, influencing handling, safety, and universal overall performance, particularly in motorsports. This research is valuable for race teams, tire manufacturers, and vehicle safety developers, helping optimize setups, format durable tires, and decorate and manage systems under diverse tire conditions.

The author of [16] explores the use of methods analyzing and data-driven techniques to measure pressure behavior, vehicle dynamics, and race technique optimization in Formula Student competitions. By analyzing onboard telemetry data, cause pressure inputs, and vehicle responses, the study employs supervised analyzing models (Random Forest, XG-Boost, Neural Networks) for general over-all performance prediction and reinforcement analyzing for driving technique optimization. Using genetic algorithms and Bayesian optimization, the research refines vehicle setup parameters along with suspension, aerodynamics, and powertrain configurations. Testing executed through simulated race environments and real-worldwide trials with Squadra Corse PoliTo examined a 5.7 percent bargain in lap times, an 8.3 percent improvement in fuel efficiency, and a 12.1 percent decrease in tire wear. The have a study highlights AI‘s functionality in refining driving techniques, optimizing race techniques, and improving vehicle general overall performance, supplying a competitive element for motorsport teams

The author in [17] investigates the usage of machine learning models to fore-cast tire strength, a key element affecting tire degradation in Formula One, which is crucial to determining the ideal time to pit stop. Its data is taken from the Mercedes-AMG PETRONAS F1 team, such as speed, throttle, brake, and guidance inputs. Among RNNs and a Transformer-primarily based total version) along baseline methods like Linear Regression and XGBoost, XGBoost tested the very best accuracy, even though deep gaining knowledge of models confirmed the capacity for additional refinement. Explainable AI (XAI) strategies have been applied to decorate version interpretability, such as function significance evaluation to discover influential enter elements and counterfactual reasons to demonstrate how to input modifications affect predictions. The findings endorse that correct tire strength forecasting can assist F1 groups in optimizing pit prevent timing and tire selection, supplying stability among predictive accuracy and interpretability. This study bridge the space among superior AI strategies and sensible motorsport applications, presenting an automated, facts-pushed framework for race method optimization.

The author of [18] examines the application of machine learning models to predict the overall performance degradation in racing due to degradation of tires. By analyzing historical race telemetry data, including speed, braking patterns, tire temperatures, and track conditions, the study develops predictive models that estimate tire lifespan and grip levels over a race stint. The research compares supervised analyzing strategies, which include regression models and neural networks, to find out the most accurate approach. The end became that optimization of tire converting method can assist with decreasing pit forestall instances that is useful for making better strategies and to enhance cars usual performance. The author highlights how AI-pushed tire control structures can improve decision-making in motorsports, decreasing the reliance on conventional heuristics and enhancing race outcomes.

The author in [19] investigates the use of tree based machine learning models for the prediction of various performance metrics of formula 1. The aim of this paper is to find ways to improve the accuracy of tree-based models with data on motorsport, which are inherently noisy and inefficient. By using models like Decision Tree the author tries to find the factors affecting the outcomes of the model.

III. METHODOLOGY

This section describes the data collection process, various features that were considered while creating the dataset, and data cleaning, followed by a brief explanation of different machine learning models used for making predictions of lap time in a race with the help of telemetry data.

A. Dataset

The dataset used in the research includes the Formula 1 data of all the races from the year 2019 to the year 2024, all the data was taken from the fastf1 library and other publicly available telemetry data. The dataset consists of data from all the weekend race days throughout the period. Each entry consists of variables like Year, Race, Round, Driver, LapNumber, LapTime, Sector1Time, Sector2Time, Sector3Time, CorneringSpeed, and CompoundUsed, which helps in correctly understanding the various factors that could affect the lap time during the race. Other external factors like track temperature, air temperature, humidity, and rain were also considered to improve the predictability of the models. A variable LapTimeDelta was engineered to show the consistency of the driver in each race, which helps in knowing what one can expect from a driver. Fig.1 illustrates the data flow in this study

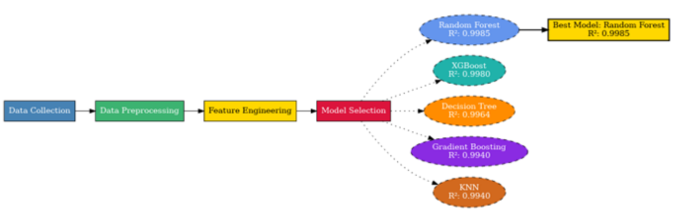

Fig. 1. Pipeline visualization

The data set obtained was cleaned and all NaN values were handled to ensure that the model was accurate while predicting lap times. The data visualization of the dataset is crucial as it helps us to find the data that has a high correlation with the target variable and to remove variables that have no impact or relation with the target variable. This, in turn, facilitates the process of narrowing down the models required for the research. Fig.2. is the correlation heatmap of all the features present in the dataset with each other. It helps us understand how influential is a feature with respect to oth-er features.

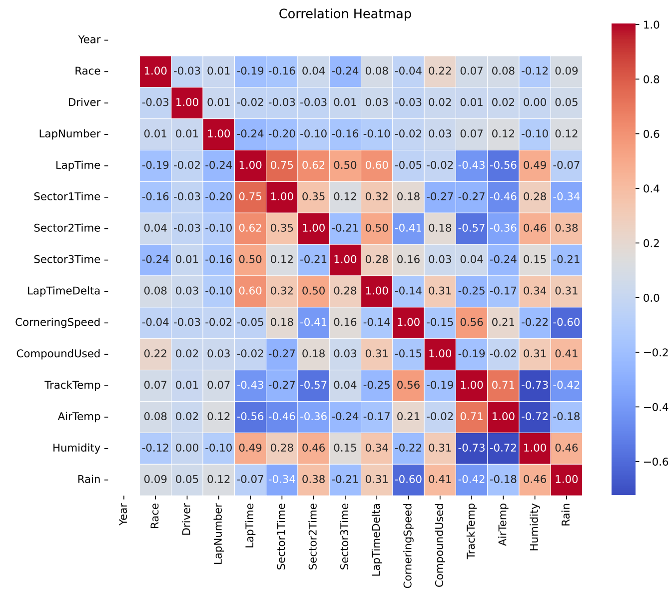

Fig. 2. Corelation Heatmap

B. Models

This paper utilizes multiple machine learning models to predict lap times in Formula 1 races. Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression (GBR), K-Nearest Neighbors (KNN), and XGBoost (Extreme Gradient Boosting) are employed, as each model emphasizes different approach-es to handling various aspects of the dataset.

*1) **Decision Tree Regressor**:* The Decision Tree Regressor is a Machine-Learning model that divides data into smaller sizes depending on various parameters and follows a tree-like structure. It is known for its robustness in capturing nonlinear relations between several features, which makes it suitable for handling multiple complex relations between track and weather data features with lap time; it showcases these abilities by having an R^2 score for the training data equal to 0.9969 and a R^2 score for the test data equal to 0.9927. Fig. 3. gives a graphical representation of the feature importance of the Decision Tree Regressor.

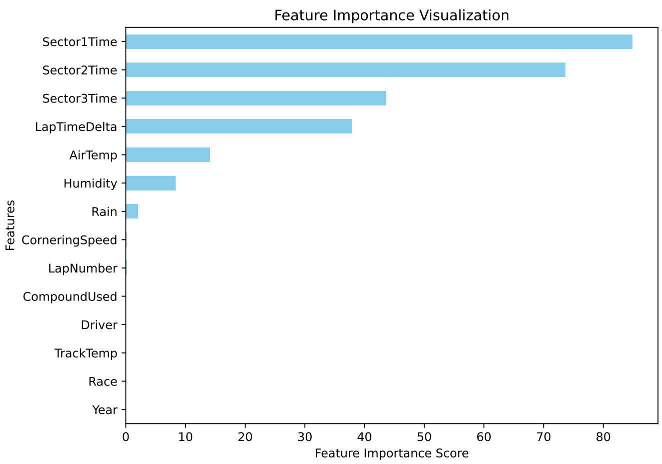

Fig. 3. Importance features

*2) **Random Forest Regressor**:* Random Forest Regressor is a powerful Machine-Learning model capable of making several decision trees that are then combined to find the average of their predictions to improve the accuracy of the predictions being made. This model helps us learn the most important feature to predict the lap time while showing an R^2 score for the training data equal to 0.9997 and an R^2 score for the training data equal to 0.9982. Fig. 4. gives a graphical representation of the feature importance of the Random Forest Regressor.

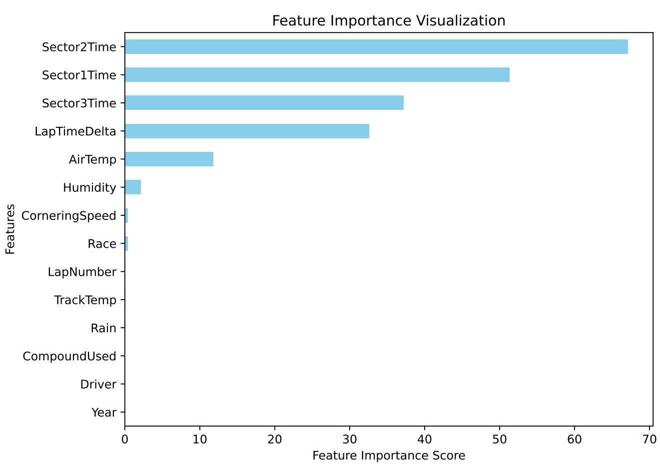

Fig. 4. Importance features

*3) **The Gradient Boosting Regressor (GBR)**:* GBR makes several weak models and focuses on correcting the error it made on the previous one to improve the accuracy of the prediction. It has an excellent ability to manage complex relations and has superior accuracy compared to other traditional models like Random Forest and decision trees. Its R^2 score for the training data is equal to 0.9931, and an R^2 score for the test data equal to 0.9921 conforms to its ability to handle complex relations accurately. Fig. 5. gives a graphical representation of the feature importance of GBR.

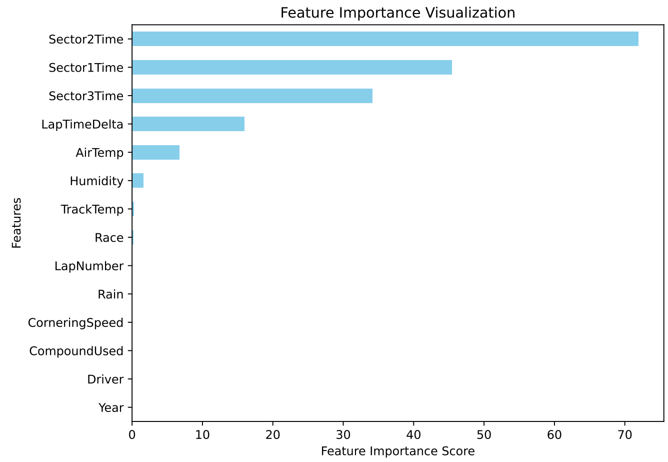

Fig. 5. Importance features

*4) **K-Nearest Neighbors (KNN)** :* KNN makes use stored training examples to build a model rather than making use of any complex mathematical function to make predictions with well known ability to predict patterns which is helpful here in the prediction of lap time that can be done based on previous race data as seen buy it R^2 score for the training data equal to 0.9959 and R^2 score for the test data equal to 0.9938. Fig. 6. gives a graphical representation of the feature importance of KNN.

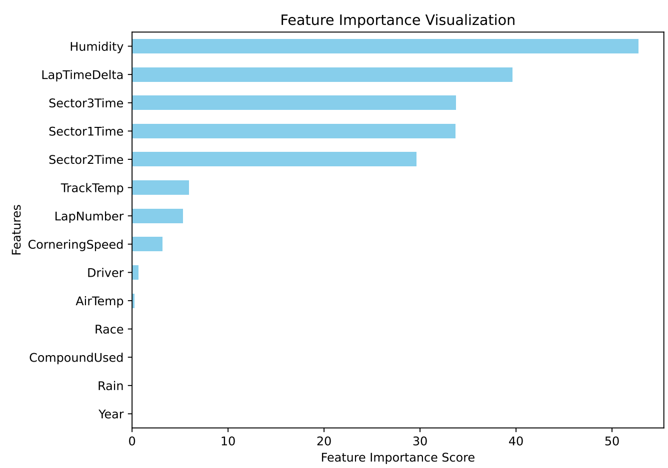

Fig. 6. Importance features

*5) **XGBoost(Extreme Gradient Boosting)**:* XGBoost, also known as Extreme Gradient Boosting is a sophisticated implementation of gradient-boosting framework where decision trees are trained and learn the differences between the predicted and test data and rectify those errors in sequential steps. It is a robust and highly efficient model because it builds trees parallelly instead of sequentially, making it useful for optimizing and analyzing existing strategies. The R^2 score for the training data is equal to 0.9995, and the R^2 score for the test data is equal to 0.9985. Fig. 7. gives a graphical representation of the feature importance of XGBoost.

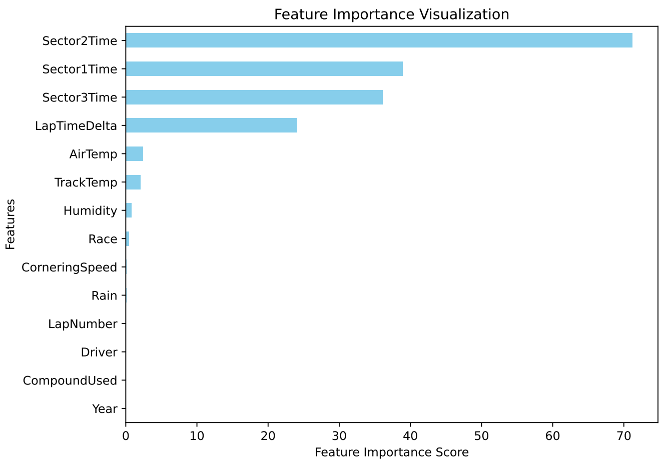

Fig. 7. Importance features

IV. RESULTS AND DISCUSSION

The primary objective of the paper is to construct and evaluate models to predict lap times in a race using machine learning. In this part of the paper, we will discuss the results obtained through the machine learning models. Table 1 shows the scores obtained by various machine learning models.

Model	R^2	MAE	MSE
Random Forest	**0.9985**	0.1678	0.3057
XGB	0.9981	0.3488	0.3924
Decision Tree	0.9964	0.3026	0.7347
Gradient Boosting	0.9940	0.6494	1.2263
KNN	0.9940	0.6494	1.2263

TABLE I
MODEL PERFORMANCE COMPARISON.

The table indicates that among all the models we used to train the data, the Random Forest Regressor is the models that fit the data in the best way possible. This is because Random Forest Regressor uses a learning algorithm capable of making several decision trees that are combined to find the average of their predictions to improve the accuracy of the predictions being made. Mathematically, the final prediction is calculated as:

$$E[y] = \frac{1}{N} \sum_{i=1}^{N} T_i(X)$$

Where N is the number of decision trees, and $T_i(X)$ represents the prediction that was calculated from the i-th Decision Tree. Here, averaging reduces the variations and generalizes the model correctly across various data features.

The main reason why this model worked the best here is because of its ability to reduce the number of variations while going through multiple trees and making sure that there is no biasing. As we know, various trees are used to get to the final predictions, and the variance reduces by:

$$\sigma^2(y) = \frac{1}{N} \operatorname{Var}(T_i(X))$$

The variance of the individual trees $Var(T_i(X))$ decreases as the number of trees N increases. This makes sure that there is some stability while we are predicting the final lap time.

The Random Forest model finds the feature importance of the entered data by itself, finding the most influential features for forecasting the lap times. This is mathematically done by reducing the variance across several trees:

$$I(x_j) = \sum_{t \in T} X_j \cdot \Delta t$$

where $\Delta_{t,j}$ represents the contribution of feature x_j to reducing error in tree t. LapTimeDelta, Track Temperature, and Tyre Compound were ranked as the most influential factors in predicting the lap times in our dataset. This helps understand factors like driver consistency and tire degradation, which are essential when making a strategy. On top of that, as it is better at handling missing and noisy datasets than many other machine-learning models, it handles the discrepancies in the publicly available data better than other models.

Model	*Dataset*	R^2	MAE	MSE
Random Forest	*With LapTimeDelta*	0.9985	0.1678	0.3057
	Without LapTimeDelta	1.0000	0.0006	0.0000
XGB	*With LapTimeDelta*	0.9980	0.3488	0.3924
	Without LapTimeDelta	0.9938	0.6312	1.0904
Decision Tree	*With LapTimeDelta*	0.9964	0.3025	0.7346
	Without LapTimeDelta	1.0000	0.0170	0.0004
Gradient Boosting	*With LapTimeDelta*	0.9940	0.6493	1.2262
	Without LapTimeDelta	0.9999	0.0385	0.0172
KNN	*With LapTimeDelta*	0.9940	0.6493	1.2262
	Without LapTimeDelta	0.9956	0.4950	0.8471

TABLE II
ABLATION STUDY FOR LAPTIMEDELTA

Table II provides a visualization of how well the data fit the models based upon the coefficient of determination (R^2), mean absolute error (MAE), and mean squared error (MSE), both with and without the inclusion of the LapTimeDelta feature during model training. When the data in the table is analysed it can be found out that the feature has really strong influence on the results. In particular, both the Random Forest and Decision Tree models achieve an R^2 score of 1.000 when LapTimeDelta is included, indicating a near-perfect fit with the training data. The nearly perfect outcomes received upon training the models on a data set in the absence of Lap-TimeDelta could refer to a strong correlation between LapTimeDelta and the target feature, but at the same time it also brings up concerns about overfitting of the data which makes the model less efficient as it doesn't produce completely accurate results. Models like XGBoost and Gradient Boosting Regressor are used to maintain prediction consistency and accuracy. This is because of their ability to generalize the data in a better manner and their low susceptibility to overfitting data. This makes them ideal for applications that require the generalization of data across various variables. Even when Random Forest has a the tendency to overfit, it is preferred when the main objective is to have highly accurate results with minimal errors. This means that under controlled conditions, the Random Forest model will give the most accurate result. However, broader application-boosting techniques like XGBoost and Gradient Boosting Regressor are recommended.

V. CONCLUSION AND FUTURE WORK

This study uses multiple machine learning models to predict Formula 1 lap times using telemetry data that is publicly available. Among them, Random Forest Regression was found to be the most effective, offering high accuracy and efficiency while maintaining low error. It handled complex datasets effectively while avoiding overfitting or data leakage. The models XGB and Decision Tree performed well in predicting the lap times in such a complex data set, while Gradient Boosting and KNN performed poorly with high Mean Absolute and Squared Error. For future works, firstly, the dataset can include more telemetry features, which will increase the complexity of the data set. This could be tackled by hybrid Deep and Machine Learning models that can also be developed to pro-duce adaptive prediction models that help with real-time decision-making. This analysis highlights the power of AI in Formula 1, paving the way for a data-centric future.

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